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Introduction to the safe dental environment, patient interview

DENT1060: Dental Practice IA



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The safe dental environment

Establishing the baseline for Dentistry

Learning objectives

- Identify risks associated with dental practice
- Define standard precautions and how they translate to the safe practice of dentistry
- Describe the process for and rationale underpinning standard precautions & infection control processes

What risks are present in the dental environment?

- Cross-infection from contaminated hands due to poor compliance with hand hygiene
- Exposure to **blood**/body fluids (saliva, blood, vomitus).
- Exposure to, and or acquisition of a vaccine **preventable disease** (hepatitis B, viral influenza, TB, etc).
- Cross infection due to poor compliance with cleaning of patient care **equipment** and/or the **environment**.
- Cross-infection due to treatment of patients/clients with symptoms of **communicable diseases**, e.g. Influenza, measles and chicken pox.
- Cross-infection due to incorrect or inappropriate cleaning, disinfection or sterilisation of reuseable **medical devices** or the re-use of medical devices marked as single-use only.
- Preventable patient or staff harm due to the **absence of an infection prevention** program.
- **Non-compliance** with established policies/procedures resulting in a preventable infection.

Infection control

Definition:

- The creation of safe healthcare environments through the implementation of evidence-based practices that minimise the risk of transmission of infectious agents (NHMRC, 2019).

What does this mean?

- To prevent and to minimise the transmission of disease-producing agents
 - Bacteria
 - Virus
 - fungi
- To prevent and to minimise this transmission between:
 - Dentist/staff to patients
 - Patients to dentist/staff

Requirements for transmission of infection

Transmission of infectious agents within a healthcare setting requires all of the following elements:

1. Causative agent (pathogen)
2. Reservoir
3. Portal of exit
4. Means of transmission
5. Portal of entry
6. A susceptible host

(NHMRC, 2019)

Pathways for transmission

Contact transmission

- Direct or indirect transmission by touch/contact with bodily substances
 - Direct -> infectious agent is transferred from one individual to another (eg patient's blood invades a practitioner's open wound)
 - Indirect -> infectious agent is transferred through contaminated object/person (eg handshake and direct contact of your hand to your mouth/wound)

Droplet transmission

- Transmission of infectious agent via respiratory droplets from an infected individual (eg coughing)

Airborne transmission

- Transmission of infectious agents via particles that remain infectious over time and distance (eg aerosol generation from dental equipment)

So what exactly do we do for infection control?

The spread of microorganisms can be reduced by:

- Adhering to good personal hygiene practices, particularly effective hand hygiene and cough etiquette
- Using PPE correctly
- Limiting surface contamination by microorganisms (e.g. disinfecting chairs, equipment, tables)
- Using disposable products where appropriate (e.g., paper towels); and
- Following risk minimisation techniques such as the use of high-volume evacuation, dental dam and pre-procedural mouth rinsing.

(Australian Dental Association, 2021)

Failure to comply with the Dental Board's guidelines may lead to an investigation by a public health regulator

- There is a risk of license restriction or suspension!

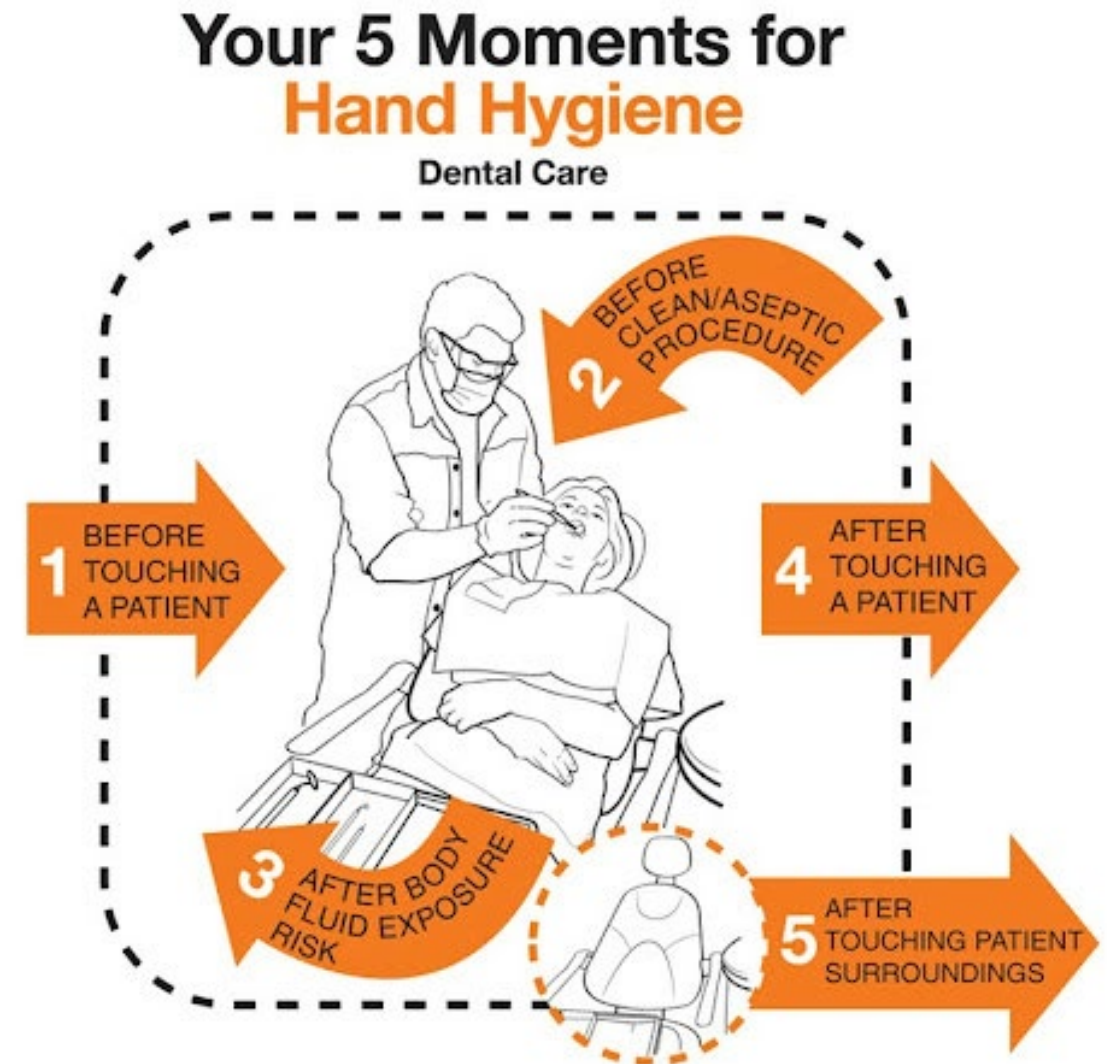
*Hand hygiene

Important rules in cleanliness

- Bare below elbows
- Proper maintenance and moisturisation of skin
 - Laceration/cracked skin allow entry of microorganism
- Open wounds require fluid-proof dressing

When are the 5 moments?

- Before seeing the patient
- During treatment (sterile treatment)
- After exposure to bodily fluids
- After discharging the patient
- After touching surroundings before disinfection



Based on "Your 5 Moments for Hand Hygiene," URL: <http://www.who.int/gpsc/5may/dental-care.pdf>, World Health Organization 2012. All rights reserved.

*Personal Protective Equipment (PPE)

Gloves

- For ALL clinical procedures, especially when coming into contact with mucous membranes

Mask

- Protects the mucous membranes of the nose and mouth against aerosols, splashes

Eye Protection

- Protection of mucous membranes of the eyes against aerosols, splatter of blood or saliva
- Prescription glasses do not replace eye protective wear, however prescription can be added into some eye protective wear

Protective clothing

- Protective gown/surgical gown, protective coats -> remove before leaving clinic premises
- If daily scrubs are to be worn -> another layer of protective clothing on top to minimise contamination

Sterilisation vs Disinfection

Sterilisation: process of rendering an item free of all forms of viable micro-organisms, including spores

- The most efficient and simplest method of sterilising equipment is pressurised steam process
- Known as autoclaving (TGA approved & operated according to AS/NZS 4187 and/or AS/NZS4815)
 - As side a note: even autoclaving cannot achieve full sterilisation, but it does remove microbial load to an acceptable level to be labelled as “sterilised”

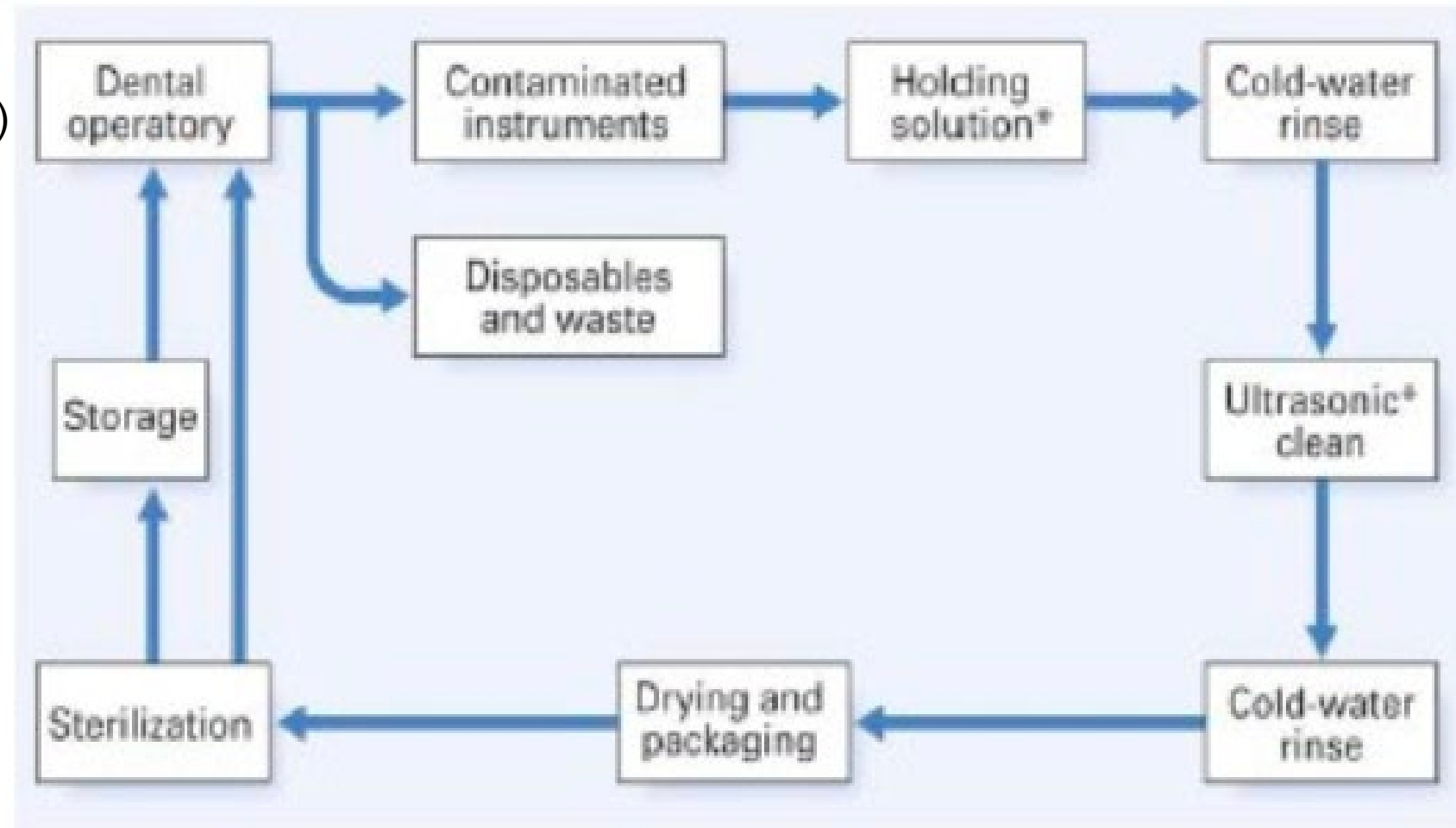
Disinfection: process of using chemicals to kill microorganisms on surfaces

- This process in itself does not necessarily fully kill all microorganisms or remove them/broken down by products from the surfaces
- Reduces the microbial load less than the sterilisation process

Why do we need to know the difference?

Process of sterilisation in Dentistry

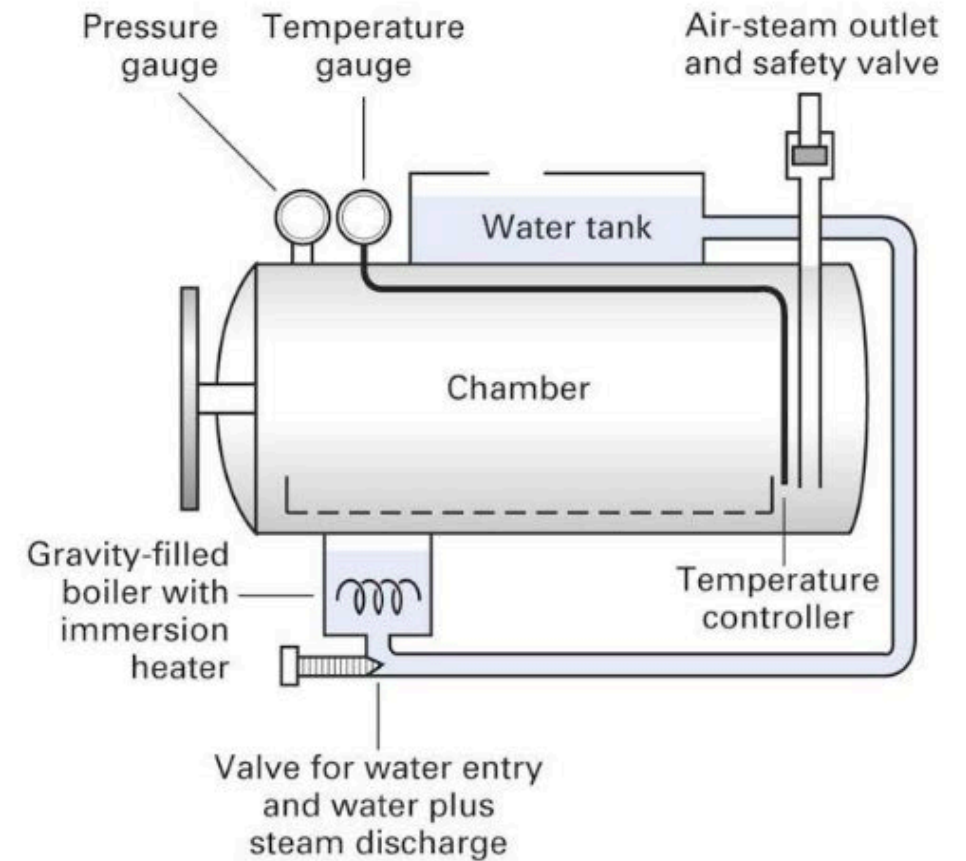
1. Preparation (PPE)
2. Cleaning (detergent solution)
3. Rinsing
4. Drying (lint free towels)
5. Packaging (steri pouch)
6. Sterilisation
7. Storage
8. Documentation



Autoclave machine

Types of sterilisation cycles:

- **N class cycles** – used for unwrapped, solid items. Steam pushes the air downwards using gravity and forces it out a port in the bottom of the chamber;
- **S class cycles** – specified by the manufacturer and used with multi-pulse vacuum steam sterilisers to suit loads of certain types and configurations; and
- **B class cycles** – for hollow objects where the ratio of the length of the hollow portion to its diameter is more than 1:5. In these cycles there is a greater challenge for air removal. Air is exhausted by a mechanical pump to create a vacuum before steam is introduced into the chamber.



Maintenance and testing

As alluded to in prior slides, as we are a healthcare profession, we are regulated by standard that we need to uphold when:

- Investigating the patient
- Performing treatment
- Maintaining/cleaning equipment

Autoclaves are subject to:

- TGA approval for the machine to be allowed to be sold and used in Australia
- Operated according to standards (Helix test and Bowie-Dick test)
- Undergo process of validation upon installation (similar with other equipment such as the xray machine)
- Regular inspection and performance testing to ensure it is still to standard
- All cycles of autoclaving must have have batch control identification

Dental unit waterline (DUWL)

The dental unit waterline is a component in the dental chair where water may remain stagnant -> fosters microbial growth!

- The growth of microbes are counted in Colony Forming Units (CFU)/mL
- The growth of these microbes can then be produces as aerosols into the patients
- Microbes can still grow in flowing dental unit waterline if the water flows slowly!

CFU requirements for different types of water

- Surgical procedures -> sterile (0 CFU/mL)
- External water (piped water) -> 20~50 CFU/mL
- Dent unit handpiece -> <200 CFU/mL
- Drink cup water -> potable water (<500CFU/mL)

Managing the waterline

- Some older dental chairs have been found with 100,000 to 1million CFU/mL in the past...
- All equipment that supplies water into the oral cavity must be fitted with non-return valves
- If the chair uses a separate bottled water supply -> chemical disinfection of water (or use sterile water if available) for the dental unit
- Minimum 2 minutes of air + waterlines flush at beginning of day
 - 30seconds between patients
- End of day (or end of week depending on chemical) chemical cleansing of suction lines



Why does this matter?

- Just within a normal dental chair, the CFU/mL can fluctuate between 0~80,000, with an average of 9,500
- Patients can be exposed to various microbes:
 - *Pseudomonas aeruginosa* (survives soap cleaning)
 - *Legionella* organisms (immunocompromised and smokers at great risk)
 - *Eschericia Coli*, and others

You don't want to be the responsible clinician if your immunocompromised patient becomes sick

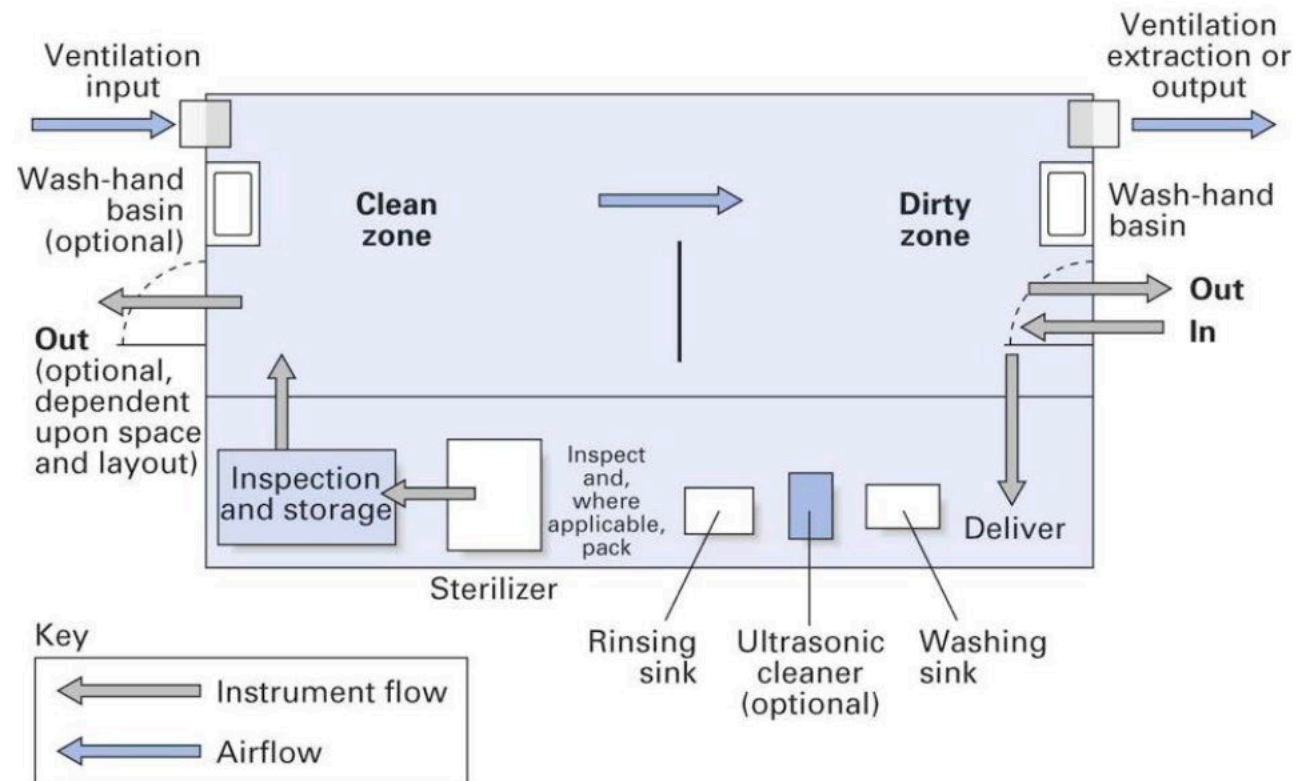
A word for warning for your career

Do **NOT** underestimate the weakness of an immunocompromised (IC) patient

- Every single disease exists in a spectrum of severity
 - Eg think about when you have a cold...
 - The severity of your cold symptoms doesn't run at 100% for the entire duration of your sickness
- One IC patient's system will not be at the same level as another
 - A single patient's immunity itself can also have times where it is lower than another day
- IE, an “immunocompromised” patient and an IMMUNOCOMPROMISED patient will tolerate different levels of exposure
 - Don't draw an assumption just because that one IC patient you've seen “was fine”

Sterile vs non-sterile (treatment room)

Although the treatment room in itself is to be considered contaminated, it is still important to identify which areas in the room are relatively more clean to appropriately distinguish clean and dirty items (and where they can be placed)



Sterile vs non-sterile (equipment)

Any instruments placed into the contaminated zone for a treatment session but not used during that session must be regarded as contaminated.

For this reason, all bulk supplies such as opened boxes of gloves, cotton rolls or gauze must be stored outside the contaminated zone and protected from contamination from splashes and aerosols.



Introduction to sharps management in surgical procedures

Key compliance points for surgical procedures

- Staff perform surgical hand preparation and wear sterile gloves.
- Staff wear an appropriate gown, and hair is controlled.
- There is a defined working field, with sterile drapes.
- Any irrigation solutions used are sterile.
- All instruments that enter tissue are sterile at the point of use, and have been wrapped, with the use of batch control identification.
- All staff involved in surgical procedures should be trained regularly in surgical asepsis and their competency assessed and performance reviewed.

Key compliance points for sharps handling and disposal

- Correct sharps handling procedures are in place.
- Sharp items are removed from the working area early in the changeover sequence.
- Approved sharps containers are used for disposal, and these are located at appropriate locations.

Introduction to sharps management in surgical procedures

Do not!

- Pass around sharps unless sharps component is protected/removed
- Resheathe needles/sharps unless with proper recapping device/appropriate handling technique
- Leave sharps exposed and unattended (resheathed appropriately or transported to sterilising room to be binned into sharps waste bin



Delegation and responsibilities

It is the clinician's responsibility to ensure that:

- Equipment is properly sterilised
- Sharps are appropriately handled and disposed of
- All attending staff also involved with the patient have appropriately disinfected/protected themselves

It is common for private dental practices to delegate this task to staff and trained nurses/assistants

- This does not make the responsibility transfer to them
- The clinician who last handles the equipment and onto the patient becomes responsible!!!

The Dental environment

The premises of the clinic is usually divided into distinct zones

- Office areas (clean zone)
 - Reception
 - Staff room
 - Waiting room, and etc
- Contaminated areas
 - Sterilisation room
 - Treatment rooms

Treatment room

- The treatment room is further divided into its relative clean and dirty zones

Exposure incident

Any incidents where a contaminated or possibly contaminated object/substance breaches the skin or mucous membrane (including contact with eyes)

- Eg blood/saliva contaminated dental instrument “pokes” the skin of practitioner

All incidents must be reported and tracked

Blood and body fluid exposure protocol

First aid

- Stop work immediately, regardless of the situation (e.g. even if administering local anaesthetic or undertaking another type of invasive procedure).
- Allow the wound to bleed and clean it thoroughly with soap and lukewarm water. There is no benefit in squeezing the wound. Do not apply disinfectants as some are irritants and retard healing.
- Flush mucous membranes/conjunctiva with normal saline or water. If contact lenses are worn, remove after flushing eye and clean as usual.
- Further management of the wound is dependent on the nature of the injury.

A full record of the incident should be made including details of:

- who was injured;
- how the incident occurred;
- type of exposure;
- presence of visible blood on the device causing the injury;
- whether a solid sharp object or hollow bore object or needle was involved;
- gauge of the needle;
- time the injury occurred;
- what action was taken;
- who was informed and when; and
- details of the patient being treated.

Mandatory testing for exposure incidents

- Baseline blood test of PATIENT (source) and injured person (staff)
 - Ideally same day of injury
 - If pt declines blood test, unfortunately you can only investigate the injured person
- Follow up review blood test
 - 3month if HIV
 - 6month if Hepatitis B or C
- If there is a high risk category for infectious transfer, may need referral to infectious disease physician

My patient is positive for Hep B/C or HIV

Your duty of care as a practitioner means that you cannot discriminately prioritise or “choose” which patients to see

- All patients must be treated the same, unless there is a genuine medical/legal/personal factor that may compromise an unbiased care to the patient

If you sustain a sharps injury:

- Follow the usual mandatory incident report protocol
- Seek counselling or further advice from the GP if you need

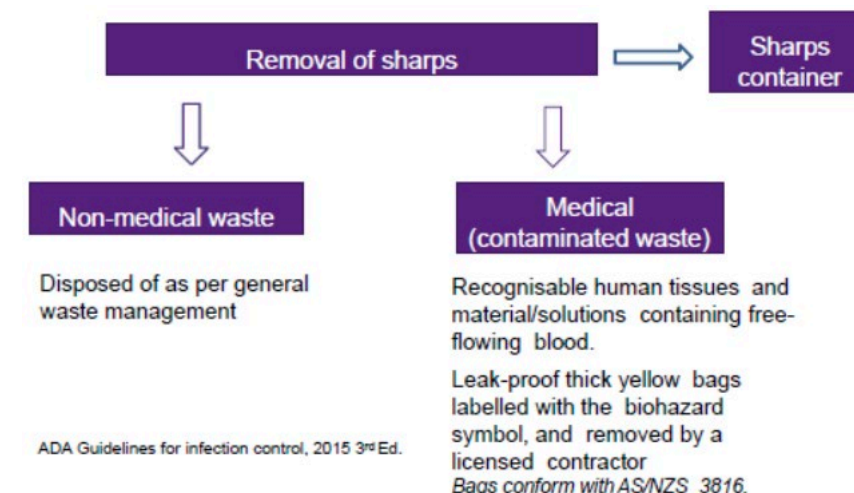
Clinical waste management

Domestic waste vs medical (contaminated) waste

- Domestic waste is treated as general waste
- Medical waste may be placed through an incineration process

IE: contaminated waste and clinical waste must be binned appropriately

- Any extracted teeth must be cleaned of visible:
 - Blood/saliva
 - Debris
 - Soft tissues
- Extracted teeth can be disposed of in domestic waste once cleaned
 - Teeth that contain amalgam restorations (grey filling) may be illegal to incinerate in some states!
 - Cannot bin in medical waste





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Ergonomics

Learning objectives

- Identify musculoskeletal risks associated with the practice of Dentistry
- Demonstrate correct ergonomics to reduce work-related musculoskeletal injuries
- Identify additional strategies to promote ergonomic practice of Dentistry

What is ergonomics?

Ergonomics is an applied science of the “fit” between the practitioner and their technological tools or environment.

- Ergonomics is not simply an understanding of common notion, but an application of demonstrated research and scientific evidence for the most ideal retention of healthy posture

The ideal ergonomics in Dentistry utilises:

- Neutral positioning
- Use of ancillary equipment
- Implemented routine self-care and rest
- Gold standard instrument grasping technique



Why is this important?

Musculoskeletal disorder is a condition where parts of the musculoskeletal system are injured over time

- Muscles, tendons, nerves

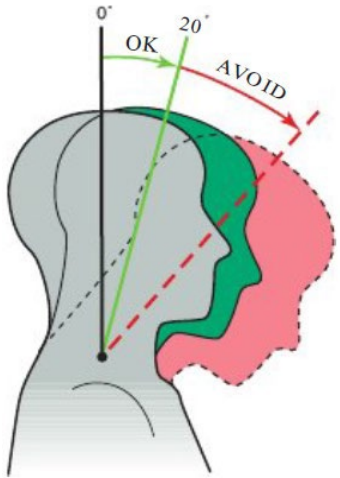
Injuries to MSD are caused by:

- Fast-paced or repetitive movements
- Forceful movements
- Poor posture

Dental literature identifies that Dentists and Dental hygienists are regularly exposed to ergonomic risks factors, leading to discomfort, pain, and disability.

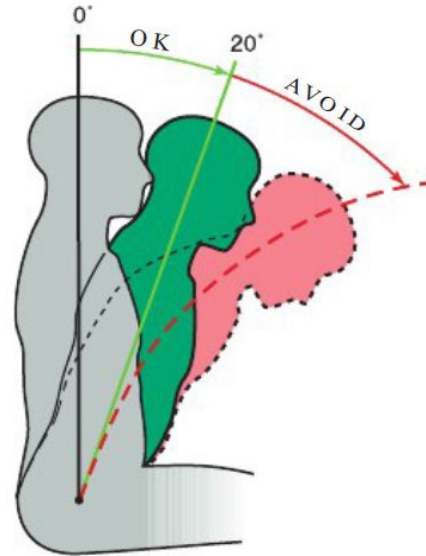
- Systematic review evidence reports that 29.5% of surveyed practitioners were forced to retire prematurely due to musculoskeletal disorder
 - Halkai 2022

Neutral position: Upper body



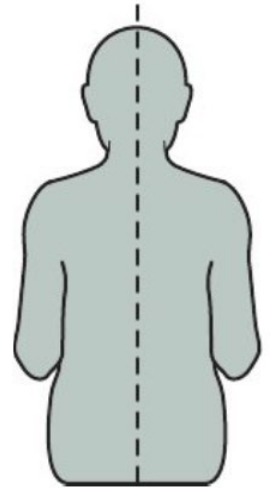
Neck

- Head tilt of 0 ~ 20 degrees



Back

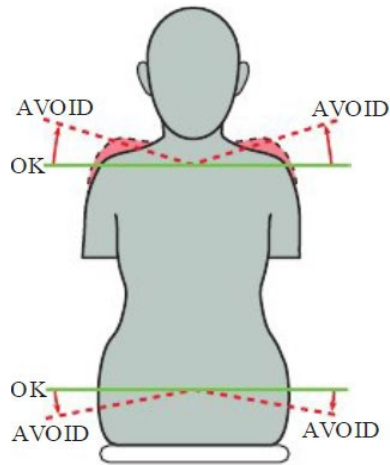
- Trunk flexion of 0 ~ 20 degrees
- Lean **slightly** forward (hinge motion)



Torso

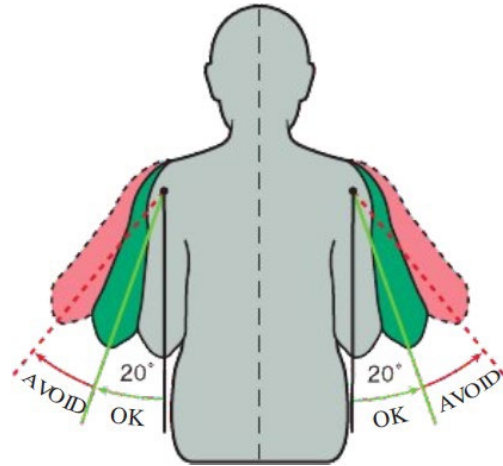
- In line with long-axis of body

Neutral position: limb extension



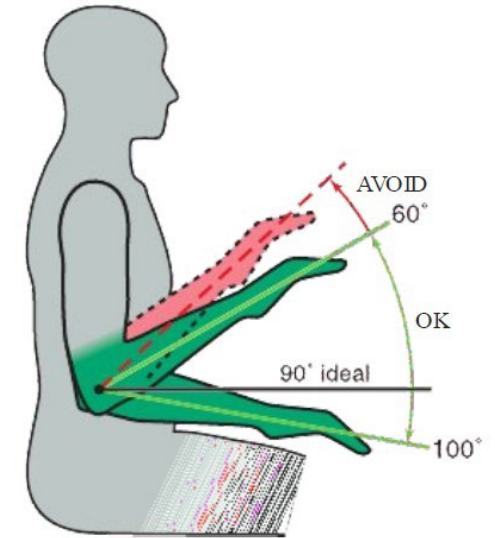
Shoulder

- Parallel to horizontal plane



Upper arm

- Arms should hang parallel to long-axis of body
- Elbows are at waist level, or near
- Elbow may be up to 20 degrees away from body



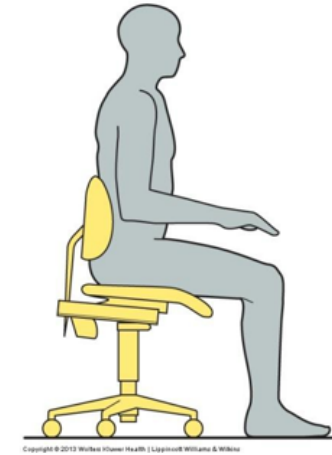
Forearm

- Parallel to floor
- If raised or lowered: by pivoting at elbow, not shoulder

Neutral position: seating

Your seating position should ideally engage your back entirely into the back-rest, and your buttocks all the way back into the chair

- Weight is evenly distributed on both hips
- Horizontal seat angulation should be adjusted to be a few cm's higher at the rear than the front
 - Seat height should be low enough for the feet to be flat on the floor, at shoulder-width apart
 - I.E. Hips are slightly higher than the knees + wide base of support with feet apart



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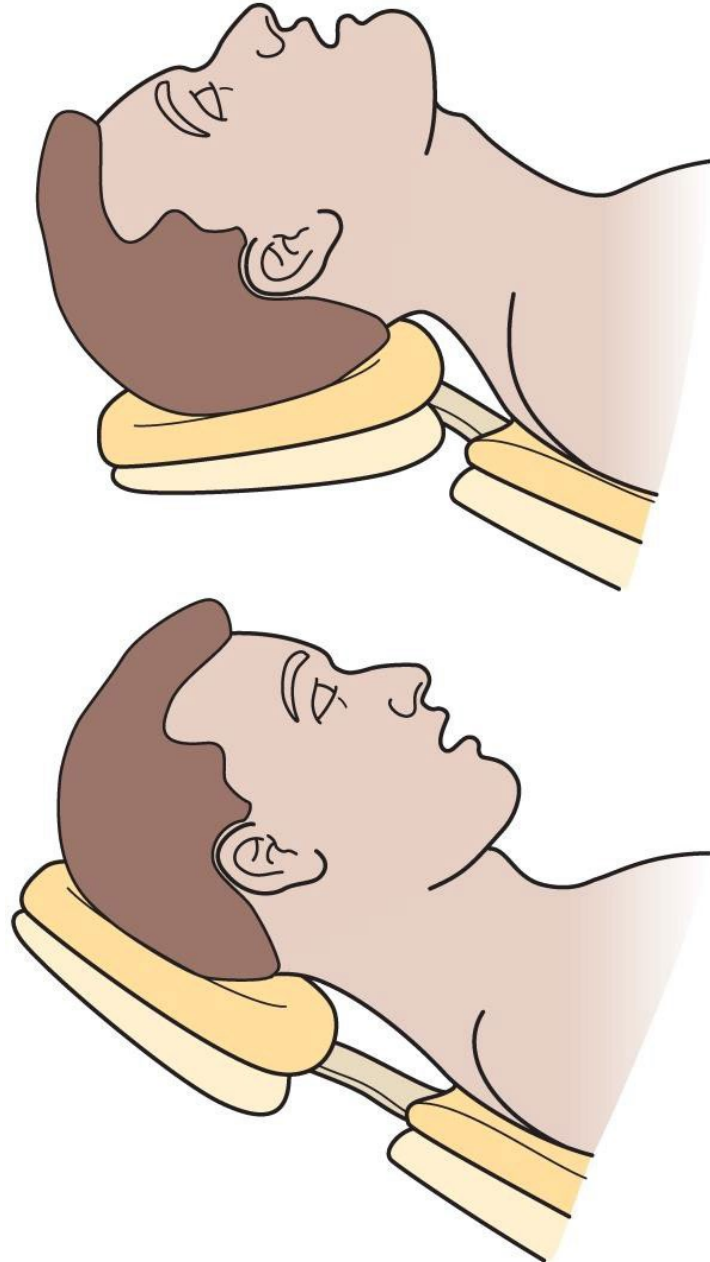


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Neutral position: patient

Dependent on whether you are working in the maxillary (upper jaw) or mandibular (lower jaw) region, the position of the patient's head require adjustment.

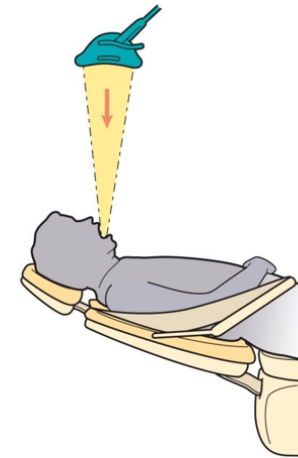
- Maxillary:
 - Nose and chin are horizontally level
 - “chin-up” position
- Mandibular:
 - Chin is lower than the nose
 - “chin-down” position



Ancillary equipment: the dental light

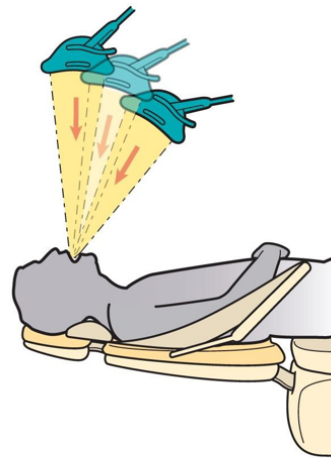
The dental light is positioned overhead such that the light beam is near parallel to the clinician's line of sight.

- Mandibular regions: dental light is positioned approximately perpendicular to the floor
- Maxillary regions: the light beam is near parallel to the clinician's line of sight **through indirect vision**



Mandibular

✓ Light directly over oral cavity



Maxilla

✓ Light Over oral cavity or patient's neck





Patient interviews

The basis of diagnosis, your official content
start to Dentistry

Learning objectives (combined with UQ extend content)

- Identify effective communication skills
 - Explain simple body language
 - Identify effective listening skills
 - Recognise barriers to effective communication
 - Distinguish between effective and poor communication styles
 - Develop a clear understanding of clinical protocols for effective patient communication, patient management and safe dental practice
 - Appreciate the rationale for thorough history-taking as an essential aspect of the comprehensive oral examination
-
- This isn't as simple as it sounds...

What is Dentistry (to the patient)?

- To receive routine care
- To address a concern
- To receive emergency care
- To receive palliative care
- Etc

Why are we "learning" how to interview the patient?

- The goal of dentistry is:
 - To investigate
 - To diagnose
 - To treatment plan
 - To provide treatment
 - To review the outcome
- Without a proper investigation of baseline results:
 - May improperly diagnose a condition
 - Cannot form an effective treatment plan addressing the cause of issue
 - May encounter "unexpected" issues that changes the protocol of management mid-treatment
 - No baseline information to compare the outcome and determine if positive or negative changes are occurring

What is involved in the patient interview?

- Creating an environment where the patient feels at ease to communicate
 - Appreciating the body language and minute gestures from the patient to direct your approach towards the patient
- Your goal is to:
 - Allow the patient to explain their history and information
 - Discern the useful and non-related informations (which may still influence your Tx planning!)
 - What “non-related” informations might there be that has a significant influence?
- Mentally and quickly correlate all information to reach a diagnosis
- Medicine places a huge importance level on this basic but key skill, there’s a reason why this is your 1st lecture into Dentistry

What information are we trying to gather?

- Pain history
 - Timing: onset, duration, periodicity
 - Location, radiation pathway
 - Quality of pain, severity
 - Relieving and aggravating factors
 - Associated symptoms
 - Taste affection, salivary flow, bruxing, clicking TMJ, ear/eye/nose symptoms
 - Other co-existing pain conditions
 - Headache, migraine, chronic widespread pain (eg fibromyalgia)
 - Impact of pain on lifestyle
- Examination of head and neck
 - Extra-oral
 - Colour changes
 - Swelling
 - Presence of skin lesions
 - Palpation of lymph nodes and salivary glands
 - Palpation of muscles of mastication
 - Palpation of sternocleidomastoid muscle
 - TMD assessment, including crepitus
 - Cranial nerves
 - Intra-oral
 - Hard tissues
 - Assessment of DMFR teeth
 - Decay
 - Missing
 - Filling
 - RCT
 - Soft tissues
 - Mucosa
 - Palate
 - Tongue
 - Floor of mouth
 - Gingival health
 - Other
 - Occlusion
 - Mouth opening
 - Presence of appliances
- General investigation
 - Family history
 - Psychological assessment
 - Medical and medication history
 - Biopsy/labworks
 - Imaging/radiographic imaging

Please don't worry, this won't be in your end of semester exam

- But this is the skill you're going to be learning over the next 5 years
- And actually also in CPDs when you graduate (believe it or not!)



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Thank you

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